# CARBOplatin AUC 4 and 5-Fluorouracil 600mg/m$^2$ with Radiotherapy

## INDICATIONS FOR USE:

| INDICATION | ICD10 | Regimen Code | HSE approved reimbursement status* |
|---|---|---|---|
| Locally Advanced Squamous Cell Carcinoma of the Head and Neck in patients who are ineligible for CISplatin based therapy | C00-C14,C30-C32, C76 | 00591a | N/A |

*This applies to post 2012 indications

## TREATMENT:

*The starting dose of the drugs detailed below may be adjusted downward by the prescribing clinician, using their independent medical judgement, to consider each patients individual clinical circumstances.*

CARBOplatin is administered on Day 1 and 5-Fluorouracil is administered on day 1-4 every 21 days for three cycles.

Chemotherapy is only to be administered if concurrent with radiotherapy.

Facilities to treat anaphylaxis MUST be present when systemic anti-cancer therapy (SACT) is administered.

| Admin. Order | Day | Drug | Dose | Route | Diluent & Rate | Cycle |
|---|---|---|---|---|---|---|
| 1 | 1 | CARBOplatin | AUC 4 | IV Infusion | 500mL glucose 5% over 30 minutes | Every 21 days for three cycles |
| 2 | 1-4 | 5-Fluorouracil[a] | 600mg/m$^2$/day | Continuous IV infusion | [a]Over 96 hours in 0.9% NaCl | Every 21 days for three cycles |
| [a]Alternatively can be administered at 600mg/m$^2$ in 1000mL 0.9% NaCl as an intravenous infusion over 22 hours on days 1-4 for a total dose of 2400mg/m$^2$ over 96 hours. | | | | | | |
| [b]See dose modifications section for patients with identified partial dihydropyrimidine dehydrogenase (DPD) deficiency | | | | | | |

Note: Administration volumes and fluids have been standardised to facilitate electronic prescribing system builds.

**CARBOplatin dose:**

The dose in mg of CARBOplatin to be administered is calculated as follows:

**Dose (mg) = target AUC (mg/mL x minute) x (GFR mL/minute +25)**



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- **Measured GFR** (e.g. nuclear renogram) is preferred whenever feasible
- **Estimation of GFR** (eGFR) can be done by using the Wright formula or using the Cockcroft and Gault formula to measure creatinine clearance
- The GFR used to calculate the AUC dosing should not exceed 125mL/minute
- For obese patients and those with a low serum creatinine, for example, due to low body weight or post-operative asthenia, estimation using formulae may not give accurate results; measured GFR is recommended
    - where obesity (body mass index [BMI] ≥ 30 kg/m$^2$) or overweight (BMI 25-29.9) is likely to lead to an overestimate of GFR and isotope GFR is not available the use of the adjusted ideal body weight in the Cockcroft and Gault formula may be considered
    - where serum creatinine is less than 63 micromol/L, the use of a creatinine value of 62 micromol/L or a steady pre-operative creatinine value may be considered
- These comments do not substitute for the clinical judgement of a physician experienced in prescription of CARBOplatin

**WRIGHT FORMULA**

There are two versions of the formula depending on how serum creatinine values are obtained, by the kinetic Jaffe method or the enzymatic method. The formula can be further adapted if covariant creatine kinase (CK) values are available (not shown).

1. *SCr measured using enzymatic assay.*

$$\text{GFR (mL/min)} = \frac{(6230 - 32.8 \times \text{Age}) \times \text{BSA} \times (1 - 0.23 \times \text{Sex})}{\text{SCr (micromol/min)}}$$

2. *SCr measured using Jaffe assay*

$$\text{GFR (mL/min)} = \frac{(6580 - 38.8 \times \text{Age}) \times \text{BSA} \times (1 - 0.168 \times \textit{Sex})}{\text{SCr (micromol/min)}}$$

Key: Sex = 1 if female, 0 if male; Age in years; BSA= DuBois BSA

**COCKCROFT-GAULT FORMULA**

$$\text{GFR (mL/min)} = \frac{S \times (140 - \text{age in years}) \times \text{wt (kg)}}{\text{serum creatinine (micromol/L)}}$$

S= 1.04 for females and 1.23 for males



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## ELIGIBILITY:

- Indication as above
- ECOG 0 to 2
- Criteria for platinum eligibility in Head & Neck squamous cell cancer should be made with reference to Szturz et al (Ref 4) with final determination made by the treating Consultant.

## EXCLUSIONS:

- Hypersensitivity to CARBOplatin, 5-Fluorouracil or any of the excipients
- Neutrophil count <1.5x$10^9$/L
- Platelet count <100x$10^9$/L
- Bilirubin >1.5xULN
- Known complete dihydropyrimidine dehydrogenase (DPD) deficiency

## PRESCRIPTIVE AUTHORITY:

- The treatment plan must be initiated by a Consultant Medical Oncologist

## TESTS:

**Baseline tests:**

- FBC, renal and liver profile
- Audiology referral as clinically indicated
- Isotope GFR measurement (preferred) or GFR / Cr Clearance estimation
- DPD testing prior to first treatment with 5-Fluorouracil using phenotype and/or genotype testing unless patient has been previously tested
    - In patients with moderate or severe renal impairment, blood uracil levels used for dihydropyrimidine dehydrogenase (DPD) phenotyping should be interpreted with caution, as impaired kidney function can lead to increased uracil blood levels. Consequently, there is an increased risk for incorrect diagnosis of DPD deficiency, which may result in under dosing of 5-Fluorouracil or other fluoropyrimidines, leading to reduced treatment efficacy. Genotype testing for DPD deficiency should be considered for patients with renal impairment.

**Regular tests:**

- FBC, renal and liver profile prior to each cycle
- Assessment of hands and feet for palmar-plantar erythrodysaesthesia (PPE) as clinically indicated

**Disease monitoring:**

Disease monitoring should be in line with the patient's treatment plan and any other test/s as directed by the supervising Consultant.



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## DOSE MODIFICATIONS:

- Consider a reduced starting dose in patients with identified partial DPD deficiency
  - Initial dose reduction may impact the efficacy of treatment
  - In the absence of serious toxicity, subsequent doses may be increased with careful monitoring
- Any dose modification should be discussed with a Consultant

### Haematological:

**Table 1: Dose modification of CARBOplatin and 5-Fluorouracil in haematological toxicity**

| ANC (x10$^9$/L) | | Platelets (x10$^9$/L) | Dose |
|---|---|---|---|
| ≥1.0 | and | ≥100 | 100% |
| 0.5-0.99 | or | 50-99 | Delay treatment until recovery |
| <0.5 | or | <50 | Delay treatment until recovery and consider reducing CARBOplatin and 5-Fluorouracil by 25% for subsequent cycles |
| Febrile neutropenia | | | |

### Renal and Hepatic Impairment:

**Table 2: Recommended dose modification in renal and hepatic impairment**

| Drug | Renal Impairment | Hepatic Impairment | |
|---|---|---|---|
| **CARBOplatin**[a] | *See below | No dose modification required | |
| **5-Flourouracil**[b] | No need for dose adjustment is expected | Mild and moderate (without renal impairment): | No need for dose adjustment is expected |
| | Haemodialysis : No need for dose adjustment is expected | Severe: | Not recommended |

[a]Renal – Content agreed by clinicians for use across NCCP regimens, Hepatic – Giraud et al 2023
[b]Renal and hepatic dose modifications as per Giraud et al 2023



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***Renal dysfunction and CARBOplatin:**

- Patients with creatinine clearance values of < 60mL/minute are at greater risk to develop myelosuppression
- In case of GFR ≤ 20mL/minute CARBOplatin should not be administered at all
- If Cockcroft & Gault or Wright formula are used, the dose should be calculated as required per cycle based on a serum creatinine obtained within 48 hrs of drug administration
- If isotope GFR is used, the dose should remain the same provided the serum creatinine is ≤110% of its value at the time of the isotope measurement. If the serum creatinine is higher than this, consideration should be given to remeasuring the GFR or to recalculating using Cockcroft & Gault or Wright formulae.

### Management of adverse events:

**Table 3: Dose Modification for Adverse Events in 5-Flourouracil**

| Adverse reactions | Recommended dose modification |
|---|---|
| Dysphagia or stomatitis<br><br>Grade 0-2<br><br>Grade 3<br><br>Grade 4 | <br><br>100%<br><br>Delay until improvement and proceed at 75-100%<br><br>Discontinue |
| Weight loss from baseline<br>≤10%<br><br>>10% | <br>100%<br><br>75% if hyperalimentation instituted, otherwise discontinue |
| Palmar-plantar erythrodysaesthesia<br><br>Grade 0-2<br><br>Grade 3 | <br><br>100%<br><br>75% |

## SUPPORTIVE CARE:

### EMETOGENIC POTENTIAL:

- As outlined in NCCP Classification Document for Systemic Anti-Cancer Therapy (SACT) Induced Nausea and Vomiting - Available on the NCCP website

**CARBOplatin**: High **(Refer to local policy).**
**5-Fluorouracil**: Low **(Refer to local policy).**

**For information:**
**Within NCIS regimens, antiemetics have been standardised by Medical Oncologists and Haemato-oncologists and information is available in the following documents:**

- NCCP Supportive Care Antiemetic Medicines for **Inclusion in NCIS** (Medical Oncology) - Available on the NCCP website
- NCCP Supportive Care Antiemetic Medicines for **Inclusion in NCIS** (Haemato-oncology) - Available on the NCCP website



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**PREMEDICATIONS:** No specific recommendations

**OTHER SUPPORTIVE CARE:**

- Anti-diarrhoeal treatment (Refer to local policy).
- Management of PPE (refer to local policy)

## ADVERSE EFFECTS:

- Please refer to the relevant Summary of Product Characteristics for details.

## REGIMEN SPECIFIC COMPLICATIONS:

**CARBOplatin**

- **Hypersensitivity:** Reactions to CARBOplatin may develop in patients who have been previously exposed to platinum therapy. However allergic reactions have been observed upon initial exposure to CARBOplatin.
- **Neurotoxicity and ototoxicity:** Neurological evaluation and an assessment of hearing should be performed on a regular basis, especially in patients receiving high dose CARBOplatin. Neurotoxicity, such as parasthesia, decreased deep tendon reflexes, and ototoxicity are more likely seen in patients previously treated with CISplatin, other platinum treatments and other ototoxic agents. Frequency of neurologic toxicity is also increased in patients older than 65 years.

**5-Fluorouracil**

- **Myocardial ischaemia and angina:** Cardiotoxicity is a serious complication during treatment with 5-Fluorouracil. Patients, especially those with a prior history of cardiac disease or other risk factors, treated with 5-Fluorouracil, should be carefully monitored during therapy.
- **DPD deficiency:** DPD is an enzyme encoded by the DPYD gene which is responsible for the breakdown of fluoropyrimidines. Patients with DPD deficiency are therefore at increased risk of fluoropyrimidine-related toxicity, including for example stomatitis, diarrhoea, mucosal inflammation, neutropenia and neurotoxicity. Treatment with 5-Fluorouracil, capecitabine or tegafur-containing medicinal products is contraindicated in patients with known complete DPD deficiency. Consider a reduced starting dose in patients with identified partial DPD deficiency. Initial dose reduction may impact the efficacy of treatment. In the absence of serious toxicity, subsequent doses may be increased with careful monitoring. Therapeutic drug monitoring (TDM) of 5-Fluorouracil may improve clinical outcomes in patients receiving continuous 5-Fluorouracil infusions.
- **Hand-foot syndrome (HFS),** also known as palmar-plantar erythrodysaesthesia (PPE) has been reported as an unusual complication of high dose bolus or protracted continuous therapy for 5-Fluorouracil.

## DRUG INTERACTIONS:

- Current SmPC and drug interaction databases should be consulted for information.



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**From 16/09/2026, this regimen will be reviewed on an 'As Required' basis. Please refer to the version control box for information, and the footer for date of last review. If you wish to suggest any updates to this regimen, please contact oncologydrugs@cancercontrol.ie**



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| Version | Date | Amendment | Approved By |
|---|---|---|---|
| 1 | 18/12/2020 | | Dr Cliona Grant |
| 2 | 27/08/2022 | Updated CARBOplatin infusion time. Updated standard wording for CARBOplatin dosing and creatinine value. Updated baseline tests. Updated dose modification of CARBOplatin in haematological toxicity.Updated emetogenic potential | Dr Cliona Grant |
| 2a | 23/11/2023 | Formatting changes and grammatical corrections. | NCCP |
| 2b | 13/03/2025 | Additional wording added to baseline testing section. | NCCP |
| 3 | 26/03/2026 | Regimen reviewed. Updated ICD-10 codes. Updated regular tests. Updated renal and hepatic dose modifications table to align with Giraud et al 2023. Updated supportive care section. Updated regimen throughout in line with NCCP standardisation. | Dr Colm Mac Eochagáin |
| 4 | 16/09/2026 | Update to indication wording Update to eligibility section.Update to review schedule. | Dr Colm Mac Eochagáin |

Comments and feedback welcome at oncologydrugs@cancercontrol.ie.



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